

Sivasankaran R

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About

Final-year B.E. Computer Science Engineering student with strong hands-on experience in backend and full-stack engineering using Java, Node.js, Python, and modern frameworks. Built scalable backend systems across AI-driven applications, blockchain platforms, and MERN-based products. Passionate about product engineering, cybersecurity, clean architecture, and solving real-world problems through reliable backend services and APIs.

Education

PSRR College of Engineering, B.E. in Computer Science and Engineering

- GPA: 8.1
- **Relevant Courses:** Data Structures, Algorithms, OS, DBMS, Computer Networks

Projects

Brain Tumor Classification using CNN-Transformer Hybrid Models (Deep Learning + Gradio) – Link

- Designed an end-to-end deep learning pipeline for automated brain tumor classification from MRI scans using CNN-Transformer architecture.
- Achieved **96% classification accuracy** on a curated MRI dataset with 3,000+ images after model tuning and augmentation.
- Integrated Grad-CAM visualizations to improve model interpretability and debugging of predictions.
- Built an interactive Gradio web interface enabling real-time inference with average response time under **1.2 seconds**.

Supply Chain Transparency Tracker (MERN + Blockchain) – Link

- Built a blockchain-based supply chain tracking system using Ethereum smart contracts to maintain immutable product history.
- Managed **500+ product records** with on-chain verification and reduced manual tracking effort by **60%** through automation.
- Developed MERN frontend with MetaMask and Web3.js enabling secure transactions with average confirmation flow under **5 seconds**.
- Integrated IPFS for decentralized document storage, reducing centralized storage dependency by **70%**.
- Designed RESTful backend APIs for product lifecycle management with role-based access and secure transaction flows.

FoodConnect – Salesforce-Based Food Donation & Distribution System (Team Project) – Link

- Designed a Salesforce-based system to manage donors, NGOs, volunteers, and distribution workflows end-to-end.
- Automated task creation and volunteer assignment using Flows and Apex, reducing manual coordination time by **55%**.
- Modeled **10+ custom objects** with Master-Detail and Lookup relationships to track donations and delivery execution.
- Built Lightning dashboards to monitor **100+ donation records** and real-time distribution status.

Technologies

Languages: Java, JavaScript (Node.js), Python, SQL, Dart, C

Frameworks, APIs & Tools: Node.js, Express.js, Spring Boot, MERN Stack, PyTorch, TensorFlow Lite, Flutter, Firebase, Git, Docker (Basics), Android Studio

Databases: MongoDB, MySQL, Firebase Realtime Database

Core Skills

Data Structures & Algorithms, REST API Development, Backend Engineering (Java, Node.js, Spring Boot), Microservices Basics, Authentication & Security, SQL & NoSQL Databases, System Design Basics, Problem Solving

Certifications

- Google Cloud Cybersecurity – Google (2025)
- Java Programming – NPTEL, IIT (2024)
- Cloud Computing – NPTEL, IIT (2025)
- Generative AI in Action – IBM (2025)